CHM_159
Worksheet_Chapter_5

Electronegativity, bond polarity, percent ionic character, Lewis structures, formal charge, exceptions to octet rule, molecular geometry

electronegativity and bond polarity

1. **Electronegativity** is the ability of an atom in a bond to draw electrons to itself. Classify the following bonds as non polar covalent, polar covalent, or ionic,

a. the CC bond in H_3CCH_3
b. the KI bond in KI
c. the NB bond in H_3NBCl_3
d. the CF bond in CF_4

2. $\mu = Q \times r$

$$\text{percent ionic character} = \frac{\mu \text{ (observed)}}{\mu \text{ (calculated assuming discrete charges)}} \times 100\%$$

1Debye(D) = 3.336 x 10-30 C.m

where C is coulombs and m is meters.

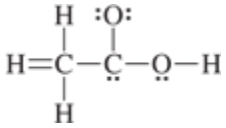
Calculate the percent ionic character of the bond in HF given that its measured dipole moment is 1.82 D and its bond length is 92 pm. The charge of an electron is 1.602×10^{-19} C and 1 D = 3.336×10^{-30} C·m.

3. Four atoms are arbitrarily labeled D, E, F, and G. Their electronegativities are as follows: D = 3.8, E = 3.3, F = 2.8, and G = 1.3. If the atoms of these elements form the molecules DE, DG, EG, and DF, how would you arrange these molecules in order of increasing covalent bond character?

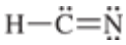
Lewis structure, resonance, formal charges

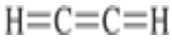
4. Draw Lewis structures for the following molecules / ions:
a. NCl_3
b. OCS
c. H_2O_2
d. CH_3COO^-
e. $COBr_2$
f. NH_4^+

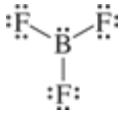
5. The skeletal structure of acetic acid shown here is correct, but some of the bonds are incorrect.
(a) Identify the incorrect bonds and explain what is wrong with them.
(b) Write the correct Lewis structure for acetic acid.

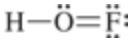


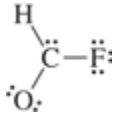
6. The following Lewis structures are incorrect. Explain what is wrong with each one, and give a correct Lewis structure for the molecule. (Relative positions of atoms are shown correctly.)

a. 

b. 

c. 

d. 

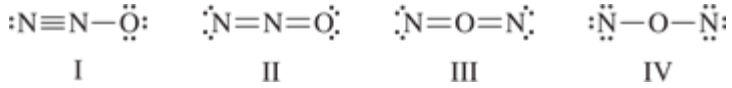
e. 

7. Draw Lewis structures for the following species, including all resonance forms, and show formal charges: a. HCO_2^-, b. HCO_3^- c. $CH_2NO_2^-$

8. Draw two resonance structures for the chlorate ion, ClO_3^- one that obeys the octet rule and one in which the formal charge on the central atom is zero.

9. Methyl isocyanate (CH_3NCO) is used to make certain pesticides. In December 1984, water leaked into a tank containing this substance at a chemical plant, producing a toxic cloud that killed thousands of people in Bhopal, India. Draw Lewis structures for NCO^-, showing formal charges. Which structure is the best one, justify your answer.

10. a. Which of the following Lewis structures are possible for N_2O?



I II III IV

b. Use formal charges to choose the best of the resonance structures shown.

exceptions to the octet rule

11. Which of the following compounds show deviation from the octet rule on the central atom: (incomplete octet, expanded octet or odd electron molecule)

BeF_2, NO_2, SF_6, PBr_5, BCl_3, XeF_4

bond energies and bond length

12. List the following compounds in order of

HCCH, H_2CCH_2, H_3CCH_3

a. decreasing C-C bond length
b. increasing C-C bond strength

13. Which of these compounds has the

H_2NNH_2, HNNH

a. longer N-N bond length
b. stronger N-N bond

VSEPR Theory

14. Complete the following table:

	Chemical Formula	Lewis Structure	Electron group geometry	Molecular geometry	Polarity
1	CS_2				
2	HCN				
3	BF_3				
4	BCl_2F				
5	SO_2				
6	SO_4^{2-}				
7	CH_2Cl_2				

8	NH_3				
9	H_2O				
10	ClO_3^-				
11	PBr_5				
12	IF_4^+				
13	IF_2^-				
14	XeF_2				
15	XeF_4				